

ISHAAN SINGH

+91 8287174260 | itzshaansingh@gmail.com | in Ishaan Singh | Singh-Ishu | Portfolio Site

EDUCATION

Vellore Institute of Technology, Chennai

B.Tech in Computer Science and Engineering, GPA: 8.64/10

Chennai, IN

Aug, 2023 - Apr, 2027

SKILLS SUMMARY

Languages: Go, Java, Python, C++, C, SQL, TypeScript

Databases & Infrastructure: Postgres, MySQL, Redis, MongoDB, Docker, Linux, Git, CI/CD (GitHub Actions)

Frameworks & Core: FastAPI, Gin, Node.js, Multithreading, Concurrent Programming

Testing & Protocols: Performance Benchmarking, Unit/Integration Testing, Load Testing, REST APIs, WebSockets, WebRTC

EXPERIENCE

ICICI Securities

SDE Intern

Delhi, IN

May, 2026 - July, 2026

- Architected an automated PR review and code-quality auditing system using Python, FastAPI, and Redis; processed GitHub webhooks with HMAC-SHA256 signature verification and orchestrated asynchronous, multi-agent linting and security scans via background queues.
- Engineered a high-concurrency validation pipeline using ThreadPoolExecutor; designed automated test suites in Pytest 100% code coverage integrated into GitHub Actions CI to validate parallel workers.

Samsung Research Institute Bangalore

PRISM Research Intern

Remote, IN

Sep, 2025 - Jan, 2026

- Architected a low-latency data ingestion pipeline utilizing Vector Databases; built a validation module to enforce schema consistency and cross-modal data integrity.
- Developed a React/FastAPI template rendering engine; designed automated API validation tests in Postman and UI tests in Cypress, validating responsive layout metrics.

Indian Oil

Software Engineer Intern

Noida, IN

June, 2025 - July, 2025

- Engineered a high-throughput backend in Go; implemented automated performance profiling and validation of real-time WebRTC telemetry pipelines to minimize memory overhead.
- Developed and validated a Scikit-learn regression model for predictive anomaly detection; designed rigorous statistical testing suites that improved leak-risk prediction accuracy and reduced false assessments by 15%.

PROJECTS

Ear Candy

Go, Postgres, TypeScript, React, Three.js, GSAP

- Architected a Go/TypeScript e-commerce backend; simulated loads up to 5,000+ writes/sec using k6 to stress-test PostgreSQL row-level locking, reducing transaction failures by 42%.
- Engineered an asynchronous state-serialization pipeline for a 3D configuration canvas; implemented integration test suites to validate frontend-to-backend state synchronization under sub-1s latency bounds, reducing payload size by 35%.

Modified AES-RSA Hybrid Encryption

C++, S-AES, Modified RSA, Multithreading, Cryptography, OpenSSL

- Designed a C++/OpenSSL hybrid cryptosystem; implemented Google Test suites validating modular key math, block padding, and decryption correctness.
- Engineered a 7-round Streamlined AES block cipher in CTR mode and integrated CRT modular reduction, boosting symmetric throughput by 4.2x and optimizing decryption latency by 35%.

ACHIEVEMENTS

Won FOSS Hack 26' out of 700+ teams, 2026

Built a low-latency client-side media pipeline by compiling FFmpeg to WebAssembly; implemented memory layout optimization and a custom runtime execution layer to bypass IPC overhead, integrating WebGPU to achieve native video rendering performance.

4th Position in BIS-Awareathon, 2025, 2025

Developed HallmarkHunters, bagging 4th Position in a hackathon hosted by the Government of India's Bureau of Indian Standards.

National LAM Research Challenge, top 10% out of 1000+ teams, 2024, 2024

Selected among top 10% nationally in algorithmic semiconductor modeling competition.

Patent Published: Autonomous Air Purifier with Smart Mobility, 2023, 2023

Application Number: 202441027661, Filed under VIT Chennai.

CERTIFICATIONS

Deep Learning Specialization – Coursera / DeepLearning.AI (Instructor/Institution: Andrew Ng) 2024

Python for Everybody Specialization – Coursera (Instructor/Institution: University of Michigan (Charles Severance)) 2023